



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-3)

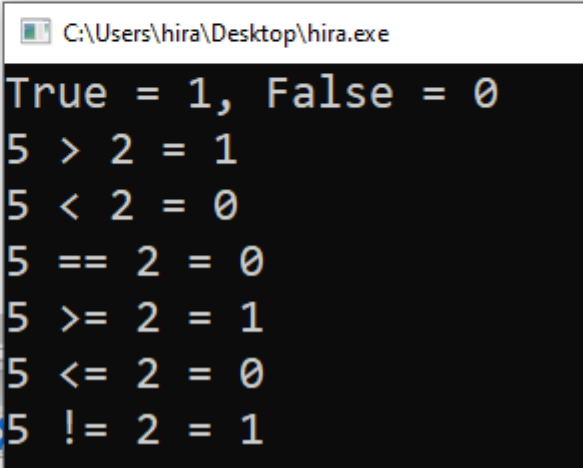
– Hira Kumar

2. Relational /Comparison Operators (6)

- Relational operators are used for checking conditions whether the given condition is true or false. If the condition is true, it will return non-zero value, if the condition is false, it will return 0.
- Example :- Suppose we have, `int X = 5, Y = 2;` then –

Operator	Name	Description	Example
>	Greater than	Check whether the left operand is greater than right operand or not.	(X > Y) will return true
<	Smaller than	Check whether the left operand is smaller than right operand or not.	(X < Y) will return false
>=	Greater than or Equal to	Check whether the left operand is greater or equal to right operand or not.	(X >= Y) will return true
<=	Smaller than or Equal to	Check whether the left operand is smaller or equal to right operand or not.	(X <= Y) will return false
==	Equal to	Check whether the both operands are equal or not.	(X == Y) will return false
!=	Not Equal to	Check whether the both operands are equal or not.	(X != Y) will return true

```
1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      int x = 5, y=2;
7      cout << "True = 1, False = 0" <<endl;
8      cout << x << " > " << y <<" = "<< (x>y) << endl;
9      cout << x << " < " << y <<" = "<< (x<y) << endl ;
10     cout << x << " == " << y <<" = "<< (x==y) << endl ;
11     cout << x << " >= " << y <<" = "<< (x>=y) << endl;
12     cout << x << " <= " << y <<" = "<< (x<=y) << endl ;
13     cout << x << " != " << y <<" = "<< (x!=y) << endl ;
14     getch();
15     return 0;
16 }
17
```



3. Increment/Decrement Operators (2)

- C++ allows two very useful operators not generally found in other computer languages.
- These are the Increment (++) and Decrement (--) operators. The operation ++ adds 1 to its operand, and -- subtracts 1.
- It is also called unary operator.
- Example:- Assume a=10; then-
 - ✓ a++ (post increment) = (10+1) = 11
 - ✓ ++a (pre increment) = (11+1) = 12
 - ✓ a-- (post decrement) = (12-1) = 11
 - ✓ --a (pre decrement) = (11-1) = 10

Solve

1. int a =7 , b =5, c= 10;
bool result = (a+b*c > 50);
2. int x = 5;
int y = ++x + x++ + --x;
3. int a =8, b=3;
bool r = ((a/b) * b ==a);

4. `int a=3;`
`int b= a++ + ++a + a--;`
5. `int a = 6, b = 9, c =2;`
`bool result = (a*b/c <= b+c);`
6. `int d =10;`
`int e= d-- - --d + ++d + d++;`
7. `int a=2, b=3;`
`bool r = (a++ *b-- > --a + ++b);`
8. `int a =4, b=5;`
`bool r = (a++ + ++b > 10);`
9. `int a=3;`
`int b= a++ + ++a + a--;`
`bool r= (b == 11);`
10. `int x=5, y=6;`
`bool new=(x++ *2 < --y +7);`

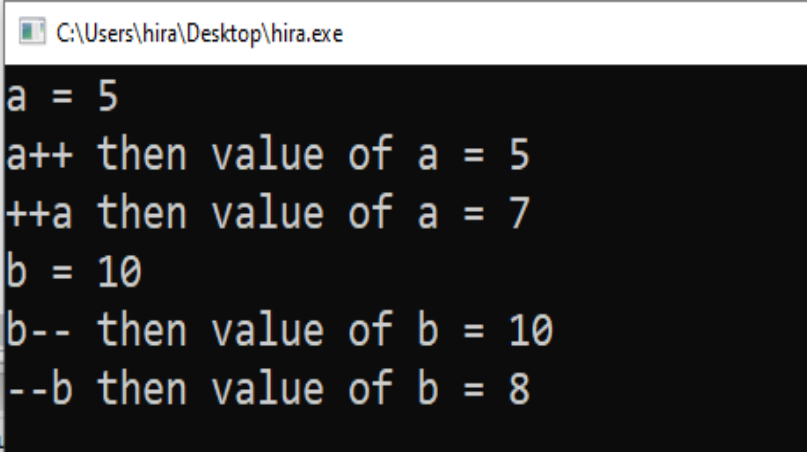
Answer

1→ 1, 2→18, 3→0, 4→ 13, 5→0, 6→20, 7→1, 8→0, 9→0,
 10→Error,(error solve answer→ 1)

```

1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      int a=5;
7      cout << "a = " <<a << endl;
8      cout << "a++ then value of a = " << a++ << endl;
9      cout << "++a then value of a = " << ++a << endl;
10     int b=10;
11     cout << "b = " <<b <<endl;
12     cout << "b-- then value of b = " << b-- << endl;
13     cout << "--b then value of b = " << --b << endl;
14     getch();
15     return 0;
16 }
17

```



The screenshot shows a C++ program being executed. The code defines two variables, `a` and `b`, and prints their values at different stages of increment and decrement operations. The output window shows the following results:

```

a = 5
a++ then value of a = 5
++a then value of a = 7
b = 10
b-- then value of b = 10
--b then value of b = 8

```

4. Logical Operators (5)

- Logical operators perform logical operations on Boolean expressions (expressions that evaluate to either true or false).
- There are three logical operators in C++:

Suppose we have,

```
int X = 5, Y = 2;
```

Operator	Name	Description	Example
&&	AND	Return true if all conditions are true, return false if any of the condition is false.	if(X > Y && Y < X) will return true
	OR	Return false if all conditions are false, return true if any of the condition is true.	if(X > Y X < Y) will return true
!	NOT	Return true if condition is false, return false if condition is true.	if(!(X > Y)) will return false

True & False

AND (&&) Table:		
A	B	A && B
T	T	T
T	F	F
F	T	F
F	F	F

OR () Table:		
A	B	A B
T	T	T
T	F	T
F	T	T
F	F	F

NOT (!) Table:	
A	!A
T	F
F	T

1 & 0

AND (&&) Table:		
A	B	A && B
1	1	1
1	0	0
0	1	0
0	0	0

OR () Table:		
A	B	A B
1	1	1
1	0	1
0	1	1
0	0	0

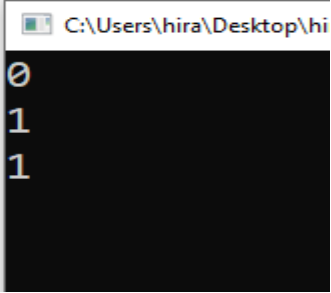
NOT (!) Table:	
A	!A
1	0
0	1

Operator	Name	Example
&&	Logical And	5 < 6 && 6 < 6 // gives 0
	Logical Or	5 < 6 6 < 5 // gives 1
!	Logical Negation (Not)	!(5 == 5) // gives 0

```

1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      int a= 5, b= 10, c=15;
7      bool r = (a>b) && (b>c);
8      bool s = (a<b) && (b<c);
9      bool t = (a>b) || (b<c);
10
11      cout << r << endl;
12      cout << s << endl;
13      cout << t << endl;
14
15
16      getch();
17      return 0;
18  }
19

```



Solve

- int a = 10, b=5, c=20;
cout << (a>b && c>a);
- int x =4;
cout << (x++ > 4 || ++x ==6);
- int a =0, b=5;
cout << !(a+b);
- int x=1, y=2;
cout << (++x && y++);
- int a=10, b=3;
cout << (a %b ==1);
- int x=0, y=5;
cout << (x++ || y++);
- int a=5, b=10, c=15;
cout << (a+b>c && b++ <c);
- int x=5;
cout << (--x && x--);
- int main()
{
 int a = 21;
 int c ;
 c = a++;
 cout << c;
 return 0;
}

Answer

1→1, 2→1, 3→0, 4→1, 5→1, 6→ 1, 7→ 0, 8 → 1, 9 → 21

=====HIRA KUMAR=====